

Python Calculator

Complete source code and quick-start guide

This handout contains a complete command-line calculator written in Python 3. It supports addition, subtraction, multiplication, and division.

Highlights

- Re-prompts for invalid numeric input.
- Validates the selected operator.
- Handles division by zero without crashing.
- Lets the user perform more than one calculation.

Run it

1. Save the source as calculator.py.
2. In a terminal, run: `python calculator.py`
3. Follow the prompts. Enter n when you are finished.

Source code follows.

calculator.py

Python 3 source code

```
1  """A small command-line calculator for the four basic operations."""
2
3
4  def get_number(prompt):
5      """Keep asking until the user supplies a valid number."""
6      while True:
7          try:
8              return float(input(prompt))
9          except ValueError:
10             print("Please enter a valid number.")
11
12
13 def calculate(first, operator, second):
14     """Return the result for a supported calculator operation."""
15     if operator == "+":
16         return first + second
17     if operator == "-":
18         return first - second
19     if operator == "*":
20         return first * second
21     if operator == "/":
22         if second == 0:
23             raise ZeroDivisionError("Division by zero is not allowed.")
24         return first / second
25     raise ValueError("Unsupported operator.")
26
27
28 def main():
29     print("Simple Python Calculator")
30     print("Available operations: + - * /")
31
32     while True:
33         first = get_number("First number: ")
34
35         operator = input("Operation (+, -, *, /): ").strip()
36         while operator not in {"+", "-", "*", "/"}:
37             operator = input("Choose +, -, *, or /: ").strip()
38
39         second = get_number("Second number: ")
40
41         try:
42             result = calculate(first, operator, second)
43             print(f"Result: {result:g}")
44         except ZeroDivisionError as error:
45             print(error)
46
47         again = input("Calculate again? (y/n): ").strip().lower()
48         if again != "y":
49             print("Thanks for using the calculator!")
50             break
51     print()
52
53
```

calculator.py

Python 3 source code

```
54 if __name__ == "__main__":  
55     main()
```